

SIMATS ENGINEERING

ASSESSMENT TOOL 3

NAME:SIVARANGJENII B
REG.NO:192521375
COURSE NAME: Computer Networks
COURSE CODE:CSA0703

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)

24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)
27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Analyze how network congestion affects packet delay and packet loss.

Network congestion occurs when too many packets travel through the network at the same time. This increases packet delay because packets wait in router queues. If the queue becomes full, packets are dropped, causing packet loss.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

- **Star Topology:** Easy to manage, failure of one device does not affect others, but failure of the central switch affects the whole network.
- **Mesh Topology:** Very reliable because multiple paths exist. Even if one link fails, communication continues through another path.

3. Why does increasing the number of routers between source and destination increase end-to-end delay?

Each router takes time to receive, process, queue, and forward packets. More routers mean more processing and queuing delays, increasing the total end-to-end delay.

4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic.

QoS prioritizes voice and video packets over normal data traffic. This reduces delay, jitter, and packet loss, providing smoother audio and video communication.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

Fiber optic cables provide higher bandwidth, faster data transmission, lower signal loss, and better resistance to electromagnetic interference than copper cables.

6. Analyze how network segmentation using VLANs improves performance and security.

VLANs divide one physical network into multiple logical networks. This reduces unnecessary traffic, improves performance, and prevents unauthorized access between departments.

7. Compare the effects of TCP and UDP on network performance for different applications.

- **TCP:** Reliable but slower because it checks errors and retransmits lost packets.
- **UDP:** Faster but unreliable because it does not retransmit lost packets. Suitable for video streaming and online gaming.

8. Why is load balancing important in large enterprise networks?

Load balancing distributes traffic across multiple servers or links. It prevents overload, improves response time, and increases network availability.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

Frequent routing updates keep routing information accurate but consume bandwidth and increase CPU usage on routers.

10. Evaluate the trade-off between network security measures and overall network performance.

Firewalls and encryption improve security but require extra processing, which may slightly increase delay and reduce network speed.

11. A network has high delay—how will you identify and fix the issue?

Check bandwidth usage, router CPU utilization, cable quality, and congestion using ping and traceroute. Upgrade bandwidth or optimize routing if needed.

12. If throughput is low despite high bandwidth, what could be the reasons?

Possible reasons include congestion, packet loss, high latency, faulty hardware, poor routing, or network errors.

13. How would you reduce packet loss in a congested network?

Increase bandwidth, configure QoS, replace faulty cables, upgrade routers, and reduce unnecessary traffic.

14. A VoIP call is breaking frequently—how would you troubleshoot?

Check bandwidth, latency, jitter, packet loss, QoS settings, and network congestion. Prioritize VoIP traffic.

15. Given a network with high jitter, suggest improvements?

Implement QoS, reduce congestion, increase bandwidth, and optimize routing paths.

16. How will you simulate a congested network in Cisco Packet Tracer?

Create multiple PCs generating traffic simultaneously using Simple PDU or Complex PDU, then observe delays and packet drops.

17. If packets are delayed in a router, what steps will you take?

Check router CPU usage, queue size, routing table, interface errors, and bandwidth utilization. Upgrade hardware if necessary.

18. How would you optimize a network for online gaming?

Use QoS, reduce latency, use wired connections, avoid congestion, and choose the shortest routing path.

19. Design a small network and explain how you will measure its performance.

Create two PCs connected through a switch and router. Measure delay using ping, throughput by file transfer, and packet loss using simulation tools.

20. If RTT is very high, how will you diagnose the issue?

Use ping and traceroute to locate delays. Check congestion, routing problems, bandwidth usage, and hardware faults.

21. Explain network performance to a non-technical person.

Network performance is how fast and smoothly information travels between devices, similar to how quickly vehicles move on a road.

22. Describe throughput using a real-life example.

Throughput is like the number of cars passing through a toll gate every minute. More cars passing means higher throughput.

23. Present how packet loss affects video streaming.

Packet loss causes buffering, poor video quality, frozen screens, and interrupted playback.

24. Explain delay types clearly in under 1 minute.

- Processing Delay: Time to process the packet.
- Queuing Delay: Waiting time in the router queue.
- Transmission Delay: Time to place data onto the link.
- Propagation Delay: Time taken for signals to travel through the medium.

25. Describe Cisco Packet Tracer in a professional way.

Cisco Packet Tracer is a network simulation software used to design, configure, test, and troubleshoot computer networks without physical devices.

26. Explain QoS in simple and technical terms.

Simple: QoS gives important data higher priority.

Technical: QoS classifies and prioritizes network traffic to reduce delay, jitter, and packet loss for critical applications.

27. How would you present your simulation results to a client?

Show the network topology, explain the testing process, present graphs and performance metrics, compare results, and provide recommendations.

28. Explain jitter using a real-world analogy.

Jitter is like buses arriving at irregular times instead of following a fixed schedule.

29. Describe how you tested network performance in a lab.

I created a network in Cisco Packet Tracer, generated traffic, measured delay, throughput, packet loss, and analyzed the results.

30. Explain the importance of performance metrics.

Performance metrics such as delay, throughput, packet loss, and jitter help evaluate network efficiency, identify problems, and improve network reliability.

31. Introduce yourself and explain your knowledge of network performance.

"My name is B. Sivarangjenii. I am an IT engineering student. I have knowledge of network performance concepts such as delay, throughput, packet loss, jitter, QoS, and network simulation using Cisco Packet Tracer."

32. Why are you interested in networking and simulation tools?

Networking connects devices worldwide, and simulation tools help design and test networks safely before real deployment.

33. What makes you confident in analyzing network performance?

I understand networking concepts, can use Cisco Packet Tracer, and know how to identify performance issues using network metrics.

34. Explain a concept you are very strong in.

I am strong in Quality of Service (QoS). It prioritizes important traffic like voice and video, reducing delay, jitter, and packet loss.

35. How would you handle a question you don't know?

I will remain calm, admit I am unsure, explain what I know, and say I will learn the correct answer.

36. Describe a situation where you solved a technical problem.

During Packet Tracer practice, I solved connectivity issues by checking IP addresses, subnet masks, and default gateway configurations.

37. Why should we select you for a networking role?

I have a strong interest in networking, understand core concepts, learn quickly, and can analyze and troubleshoot network problems effectively.

38. What challenges did you face while learning Cisco Packet Tracer?

Initially, configuring IP addresses, routing, and troubleshooting connectivity was difficult, but regular practice improved my skills.

39. How do you improve your technical skills regularly?

I practice network simulations, learn new concepts, watch tutorials, read technical materials, and solve networking problems regularly.

40. Demonstrate confidence while explaining a complex concept simply.

"Quality of Service is like giving an ambulance priority on a busy road. Important traffic such as voice and video is allowed to move first, ensuring smooth communication even when the network is busy."